Operation Ember Ascent: Mustafar Flight Report

Team Clock_it! | PS1 OpenRocket Extraterrestrial Flight Design Challenge | Body material: [FILL IN: Cardboard / PVC / Aluminium - the brief says this must be stated]

1. What we built

We made a two-stage sounding rocket in OpenRocket. It is 60.5 cm long with a 7 cm maximum diameter, and it weighs 2329 g without motors and 3528 g with them, so it is nowhere near the mass limit for any of the three body materials. The motors are an I120-15 and an I445-6. Both are I-class, and an I-class motor tops out at 640 N·s, so the combined impulse is at most 1280 N·s, comfortably under the 2800 N·s limit. The payload mass is [FILL IN: payload mass in kg, must be 0.95 to 1.05].

Recovery. Each stage has its own parachute, and both are set to deploy at apogee with a 0 s delay. Chute diameters: [FILL IN: Parachute1 and Parachute2 diameters in cm]. **Stability.** The brief doesn't set a minimum, so we are just reporting it. At launch the CG is at 28.9 cm and the CP at 37.4 cm (at Mach 0.3), which is 1.2 calibres, or 13.9% of the rocket length. Over the whole flight it ranges from [FILL IN: min and max stability from the stability vs time plot] calibres.

2. Setting up the Mustafar environment

OpenRocket doesn't have a gravity box in the launch conditions, so gravity is set with the *Planetary Environment (Mustafar)* extension on the simulation, which fixes it at 6.50 m/s^2. That extension does not touch the atmosphere, so we typed pressure and temperature in by hand. We left “Use International Standard Atmosphere” unticked and let OpenRocket work out the density itself (about 0.373 kg/m^3, matching the 0.374 in the brief).

Parameter	Earth (comparison run)	Mustafar (mission)
Gravity	9.81 m/s^2	6.50 m/s^2 (extension)
Pressure	101.3 kPa	75 kPa (750 mbar)
Temperature	15 °C (288 K)	700 K (426.85 °C)
Air density	1.225 kg/m^3	0.374 kg/m^3
Wind	[FILL IN: Earth wind setting you used]	10 m/s average at 270°, 1 m/s std dev, 10% turbulence
Launch site	0 m altitude	0 m altitude, 200 cm rod, 0° angle (vertical)

3. Do we meet the requirements?

Requirement	Our result (Mustafar)	Met?
Apogee 3900 to 4100 m	3913 m	Yes (only just, 13 m above the lower limit)
Payload 0.95 to 1.05 kg	[FILL IN: payload mass]	[FILL IN: yes/no]
Length at most 220 cm	60.5 cm	Yes
Total impulse at most 2800 N·s	Two I-class motors (at most 1280 N·s combined)	Yes
Deploy at apogee, parachute recovery	Both chutes at apogee, 0 s delay, both stages recovered	Yes
Descent / ground hit velocity 4 to 8 m/s	9.16 m/s	**No** (see section 5)

4. Mustafar vs Earth

The brief says to treat each environmental effect as a hypothesis and test it by flying the same design in both atmospheres. We flew the exact same rocket (same motors, same chutes) once on Mustafar and once with Earth settings. Here is what OpenRocket gave us.

Result	Mustafar	Earth
Apogee	3913 m	2115 m

Result	Mustafar	Earth
Time to apogee	33.9 s	20.6 s
Max velocity	246 m/s (Mach 0.439)	202 m/s (about Mach 0.59, calculated using 340 m/s)
Max acceleration	149 m/s^2	145 m/s^2
Velocity off the rod	21.8 m/s	22.3 m/s
Velocity at deployment	32.9 m/s	6 m/s
Flight time	1317 s	964 s
Ground hit velocity	9.16 m/s	2.93 m/s
Stability at launch	1.2 cal (at Mach 0.3)	1.2 cal (same, since the geometry is the same)
Downrange drift	[FILL IN: drift in m]	[FILL IN: drift in m]

One honest caveat first. Ideally we would change one thing at a time (just gravity, then just air, then just wind). We only did the full Mustafar vs full Earth comparison, so below we can say whether each prediction is *consistent* with the numbers, but we can't separate the effects cleanly. [FILL IN: if you run the one-at-a-time versions, put them here and update the table]

Change	Our prediction	What we saw	Did it hold?
Lower gravity (6.5 vs 9.81)	Weight is about 34% lower, so the rocket climbs higher and for longer. Max acceleration should rise by about 3.3 m/s^2 (that is just the difference in g).	Apogee 2115 m to 3913 m (about 1.85 times). Time to apogee 20.6 s to 33.9 s. Max acceleration 145 to 149 m/s^2, a rise of about 4.	Yes. The acceleration rise matches almost exactly. The apogee gain also includes the air, so it isn't gravity alone.
Thinner air (0.374 vs 1.225 kg/m^3, 75 vs 101.3 kPa)	Drag scales with density, so it is about 3.3 times lower. The rocket should go faster and higher.	Max velocity went from 202 to 246 m/s with the same motors.	Yes (together with gravity).
Hotter air (700 K vs 288 K)	Speed of sound goes as the square root of temperature, so it is about 1.56 times higher. The same speed should give a lower Mach number.	Mustafar max Mach 0.439 at 246 m/s, which works out to a sound speed of roughly 560 m/s. On Earth, 202 m/s is about Mach 0.59.	Yes. Mustafar flew faster but at a lower Mach number. The sound speed ratio (about 1.6) is close to the 1.56 predicted.
10 m/s side wind	The rocket weathercocks into the wind, drifts downrange, and arrives at apogee still moving sideways.	Velocity at deployment is 32.9 m/s on Mustafar vs 6 m/s on Earth, and Mustafar flight time is longer (1317 s vs 964 s). Drift: [FILL IN: drift numbers]	Partly. The deployment speed supports it, but we didn't isolate wind, and [FILL IN: whether Earth run had wind].
Thinner air and lower gravity on the parachute	Terminal speed goes as the square root of (g divided by density), so the same chute should come down about 1.5 times faster on Mustafar.	Ground hit velocity 2.93 m/s on Earth vs 9.16 m/s on Mustafar, about 3.1 times.	Direction yes, size no. See section 5.

5. The ground hit velocity: we didn't get it into 4 to 8 m/s

Being straight about this: we did not manage to get the ground hit velocity into the 4 to 8 m/s window, even after trying many times. Our best run has 9.16 m/s, and across all our attempts the number sat between about 9 and 12 m/s. We resized both parachutes again and again, and changed the motors to pull the apogee back into range when the lower gravity pushed it too high. Nothing we changed brought the number under 8.

Our best guess is that the 10 m/s side wind is the problem. OpenRocket's ground hit velocity is the total speed, so it includes the sideways speed the rocket picks up while drifting under the chutes, and that can't go much below the wind speed however big the chute is. The vertical sink rate alone is probably lower than 9.16 m/s. The Earth run backs this up: with the exact same chutes it came down at only 2.93 m/s, and the formula says the vertical speed should only be about 1.5 times higher on Mustafar, but the total came out about 3.1 times higher. So the rest is probably sideways speed. We want to be clear that this is a hypothesis, not something we proved. [FILL IN: if you plotted vertical velocity, write the vertical descent rate for each piece here]

The Mustafar run also still shows 2 OpenRocket warnings, and the speed at deployment (32.9 m/s) is high, which is rough on the parachutes. We left these in rather than hiding them. If we had more time, we would fix the sideways speed at apogee (a longer launch rod and more stability margin to cut the weathercocking) and then re-tune the chutes.

6. Conclusion

On Mustafar our rocket reaches 3913 m, which is inside the 3900 to 4100 m window, and every requirement we can check is met except the ground hit velocity (9.16 m/s against 4 to 8 m/s). Comparing it with Earth, the predictions about gravity, thinner air and temperature all held in direction and roughly in size. The wind and parachute predictions held in direction but we couldn't pin down the size without separate runs. The biggest thing we learned is that on a windy world the numbers OpenRocket reports can mix up sideways and downward speed, and that you have to look at the vertical part separately before blaming the parachute.